

Suryansh

A demo that cannot fail

No GPU

Stage 2 PoC - 1-2 Sept

SIH26104 - Team Sonix

Your mission: Make sure nothing on stage depends on luck. You do not need a GPU for any of this.

THE SITUATION - READ THIS FIRST

Where we are. SONIX listens to a call in 4-second chunks and shows green / amber / red. On the academic benchmark it is strong: **1.49% EER** on ASVspoof-2019 LA. But on our own real recordings it wrongly flags roughly **half of genuine voices as fake**.

Why. The model only ever learned from clean studio recordings of real people - same good microphone, same quiet room. Our phone and laptop recordings sound nothing like that. So the model thinks: "this is not like any real voice I know, therefore fake." **The model is not broken. It is narrow.** It has never heard normal life.

The fix - three things. (1) **Show it more kinds of real:** add genuine recordings from many phones, mics, rooms and languages to training. (2) **Make it ignore silence:** quiet gaps are what break the "minimal voice" clips. (3) **Reset the alarm levels:** our amber/red numbers came from the clean test, so we re-set them using real recordings.

HARD RULE: we never train on In-the-Wild or on any eval set. Those are our honest test. Training on them would destroy the one number that proves we generalise.

THE PLAN - THREE LANES RUNNING AT ONCE

LANE 1 - FIX THE MODEL

Silence gate, more real audio, RawBoost augmentation, retrain, recalibrate. This is what stops the false alarms.

Yugal - Akshat - Yukti

LANE 2 - HONEST NUMBERS

2021-DF EER, In-the-Wild EER, codec 2x2, metrics + calibration. These are the numbers we quote to the jury.

Navya - Anubhav - Yukti

LANE 3 - A DEMO THAT CANNOT FAIL

Replay mode on validated clips, clip library, pitch, Q&A, backup video. The demo never depends on live luck.

Suryansh - Anubhav - Akshat

YOUR TASKS

P1 TONIGHT

Add Replay (precomputed) mode to app.py

- Right now the demo runs the model live on whatever audio is uploaded - and the model currently misfires on real audio. That is a stage risk we remove entirely.
- Add a third scoring mode that reads a saved list of scores from JSON and streams them into the existing timeline and risk band at the same 0.5s cadence.
- The UI code barely changes - you are swapping where the numbers come from.

```
scores\real_01.json
{ "clip": "real_01.wav", "scores": [0.02, 0.03, 0.05, ...] }
# one score per 4s window at 0.5s hop
```

Done when: A saved clip replays exactly the same way every time.

P1
TOMORROW

Clip library dropdown

- A dropdown of validated demo clips with their known verdicts, so the demo runs off a trusted set instead of an arbitrary upload.
- Keep the file uploader too, but the jury demo uses the library.

Done when: Dropdown lists the validated clips and loads each one.

P2
TOMORROW

Keep the model tabs honest

- Tabs: Baseline / Augmented / Robust (Yugal will add head_robust.pt).
- Show the thresholds in the sidebar so the jury sees they are configurable - that is a selling point from our impact slide.

Done when: Tabs and thresholds visible and correct.

P2
TOMORROW

Record the backup video

- Screen-record a full successful run: upload, stream, green on a real clip, red on a cloned clip.
- If the laptop dies on stage, this video IS the demo. Do not skip it.

Done when: A 2-3 minute clean recording saved and shared.

YOUR CHECKLIST

- ☐ Replay mode working
- ☐ Clip library dropdown
- ☐ Model tabs + thresholds shown
- ☐ Backup video recorded

Note: You need the score JSONs from Yugal. Chase him if they are not there by tomorrow morning.

WORDS WE KEEP USING

EER	One number for how wrong the model is. Lower = better. 1.49% is strong. 50% = coin flip.
bonafide	A real human voice.
spoof	An AI-cloned / synthetic voice.
embedding	The 1024-number fingerprint of a 4-second audio chunk.
head	The small model we actually train. The big wav2vec2 front-end is frozen (never trained).
VAD	Voice Activity Detection - is anyone actually speaking in this chunk?
augmentation	Making altered copies of training audio (phone-compressed, noisy) so the model sees variety.
calibration	Making the 0-1 score mean a real probability, so thresholds are meaningful.
domain shift	Test audio looks different from training audio. THIS is our core problem.